

Explainer: Problem 5 (Rudin 7.13) — Helly’s Selection Theorem

What’s going on?

We have a sequence of functions $f_n : \mathbb{R} \rightarrow [0, 1]$, each monotonically increasing. From this sequence, we want to extract a subsequence f_{n_k} that converges *pointwise everywhere*. Part (b) upgrades to uniform convergence on compact sets if the limit is continuous.

This is an “infinite-dimensional” version of Bolzano–Weierstrass.

The big picture: a diagonal argument

- Step 1:** List the rationals $r_1, r_2, r_3, \dots$. Since $f_n(r_1) \in [0, 1]$ for all n , Bolzano–Weierstrass gives a subsequence converging at r_1 .
- Step 2:** From that subsequence, extract a further subsequence converging at r_2 . (It still converges at r_1 !) Keep going... diagonal trick: the k th subsequence converges at $r_1, \dots, r_k$. Take the diagonal to get convergence at *all* rationals.
- Step 3:** Use ~~monotonicity~~ to extend from $\mathbb{Q}$ to all of $\mathbb{R}$.
- Step 4:** This is the key step where the monotone assumption does real work.

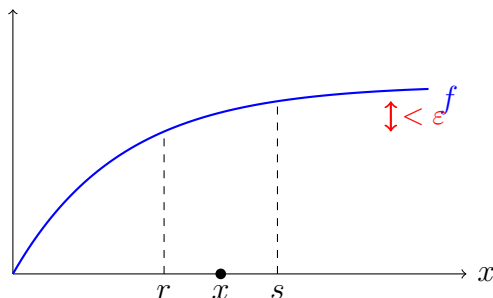
Why monotonicity matters for Step 4

Suppose we know $f_{n_k}(r) \rightarrow g(r)$ for all rationals r . Define $f(x) = \sup\{g(r) : r \leq x, r \in \mathbb{Q}\}$.

At a point x where f is **continuous**: for any $\varepsilon > 0$, pick rationals $r < x < s$ with $f(s) - f(r) < \varepsilon$. Then

$$f(r) \leq f_{n_k}(r) + \varepsilon_k \leq f_{n_k}(x) \leq f_{n_k}(s) \leq f(s) + \varepsilon_k$$

(using monotonicity of f_{n_k}), and squeezing gives $f_{n_k}(x) \rightarrow f(x)$.



Because f_{n_k} is increasing, knowing its values at rationals on both sides of x **traps** $f_{n_k}(x)$ in a small interval. Without monotonicity, knowing f_{n_k} at rationals tells you nothing about nearby irrationals.

Dealing with discontinuities of f

The limit f might be discontinuous (monotone functions can have at most countably many jumps). At jump points, the “squeeze from rationals” argument fails because $f(r)$ and $f(s)$ don’t get close.

But there are only countably many jumps! So apply Bolzano–Weierstrass one more time: from the subsequence that converges at all continuity points, extract a further subsequence converging at all (countably many) discontinuity points.

Part (b): upgrading to uniform convergence

If f is continuous, there are **no** discontinuities to worry about, and the squeeze argument works at every point. On a compact set $[a, b]$, the continuity is uniform, so the squeezing can be made uniform $\Rightarrow$ uniform convergence on $[a, b]$.

Proof outline

1. Enumerate rationals, use diagonal argument to get subsequence converging on $\mathbb{Q}$.
2. Define $f(x) = \sup\{g(r) : r \in \mathbb{Q}, r \leq x\}$.
3. Monotonicity $\Rightarrow$ convergence at all continuity points of f .
4. Extract further subsequence for (countably many) discontinuity points.
5. For (b): if f continuous, the squeeze is uniform on compact sets.